6TH GRADE MATH · UNIT 6: GEOMETRY

Dream Bedroom Makeover

Use area, surface area, and volume to redesign a bedroom on a budget.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A TV show is giving a student a bedroom makeover, and your design team is in charge. You need new flooring, a rug, paint for the walls, and storage boxes, and everything must fit the budget.

You will use area, surface area, and volume to measure the room, plan materials, and present your design.



Room Size

12 ft by 10 ft

walls 8 ft tall

Budget

$900 total

Learning Goals

- I can find the area of rectangles, triangles, and composite shapes.
- I can use a net to find surface area.
- I can find the volume of a rectangular prism.

Project Timeline

When	What you will do
Day 1	Floor Plan
Day 2	Paint the Walls
Day 3	Storage & Budget
Final	Makeover Design Board + reflection

Part 1: Floor Plan

Day 1 · The room is a rectangle 12 ft long and 10 ft wide.

Makeover Price Cards

Wood Flooring	Carpet	Paint
$3.50 per sq ft	$2.25 per sq ft	$32 per gallon covers 350 sq ft

Shag Rug	Storage Box	Wall Decal
$4 per sq ft	2 ft × 1.5 ft × 1 ft $18 each	$0.75 per sq in

1. Find the area of the floor.

   Starter: $A = l \times w =$ ___ × ___

2. Find the cost to cover the floor with carpet.

3. A triangle rug has a base of 6 ft and a height of 4 ft. Find its area.

4. An L-shaped desk top is made of a 5 ft by 2 ft rectangle and a 3 ft by 2 ft rectangle. Find its area.

Part 2: Paint the Walls

Day 2 · The walls are 8 ft tall.

5. Find the area of one long wall (12 ft by 8 ft) and one short wall (10 ft by 8 ft).

6. Find the total area of all four walls.

7. One gallon of paint covers 350 sq ft. How many gallons do you need? What does it cost?

8. Draw the net of a storage box (rectangular prism) 2 ft by 1.5 ft by 1 ft.

Part 3: Storage & Budget

Day 3 · Use the Makeover Price Cards.

9. Find the volume of one storage box.

10. You buy 3 boxes. What is their total volume and total cost?

11. Add up your carpet, paint, rug ($4 per sq ft), and boxes. Is it under the $900 budget?

Final Product: Makeover Design Board

Your final product must include:

- A floor plan with the dimensions
- Area of the floor and the rug
- Wall area and paint needed
- Volume of your storage
- Price list and total cost

Reflection

R1. Why is area measured in square feet but volume in cubic feet?

R2. How did the net help you find surface area?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Areas, surface areas, and volumes are computed correctly and used to choose materials within the budget. Goes beyond the minimum with a creative, realistic choice.	Areas, surface areas, and volumes are computed correctly and used to choose materials within the budget.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Design board shows the floor plan, calculations, materials list, and total cost. Polished, organized, and visually engaging.	Design board shows the floor plan, calculations, materials list, and total cost. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.